

EXPLAINED EXAMPLE 10 – TURBOJET ANALYSIS

- ① A turbojet aircraft engine at altitude ($T=230$ K and $P=18700$ Pa) is flying at 250 m/s with a mass flow of 680 kg/s through the engine. The inlet diffuser slows the flow down to a velocity of 125 m/s. The pressure ratio of the compressor is 25 and the isentropic efficiency of the compressor is 0.9 . Heat is added at a rate of 200 MJ/sec. What is the temperature of the air coming out the back of the turbine? Assume $c_p=1001$ J/kg-K and $\gamma=1.4$.

Given : $T_1 = 230$ K

Find : T_5

Assume : ideal gas

② $P_1 = 18700$ Pa

$V_1 = 250$ m/s

$\dot{m} = 680$ kg/s

$V_2 = 125$ m/s

$P_3/P_2 = 25$

$\eta_c = 0.9$

$\dot{Q}_{in} = 200$ MJ/s

① : steady flow

: nozzle : $\Delta p_e = \dot{Q} = \dot{W} = 0$

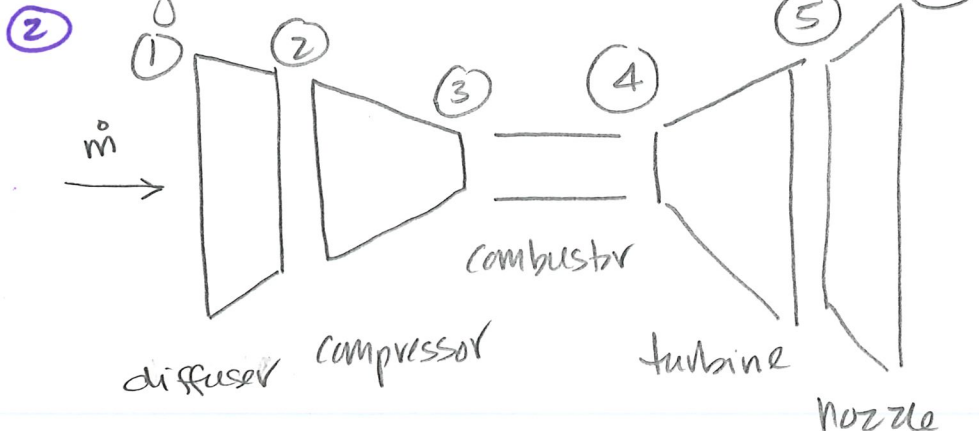
comp : $\Delta ke = \Delta p_e = \dot{Q} = 0$

comb : $\Delta ke = \Delta p_e = \dot{W} = 0$

turb : $\Delta ke = \Delta p_e = \dot{Q} = 0$

: ideal cycle $\rightarrow$ reversible

$\Rightarrow$ diagram



State 1 : $T_1 = 230$ K

① $P_1 = 18700$ Pa

$V_1 = 250$ m/s

State 2 : diffuser: $\dot{m} (\Delta h + \Delta ke + \cancel{\Delta pe}) = \dot{Q} - \dot{W}$

②

$$\Rightarrow \Delta h + \Delta ke = 0$$

$$c_p (T_2 - T_1) + \frac{1}{2} (v_2^2 - v_1^2) = 0$$

$$T_2 = T_1 - \frac{1}{2c_p} (v_2^2 - v_1^2) = 0$$

$$= 250 - \frac{1}{2002} (250^2 - 125^2) \rightarrow \boxed{T_2 = 226.6 \text{ K}}$$

①

State 3s: Isentropic compression

$$T_{3s} = T_2 \left(\frac{P_3}{P_2} \right)^{\frac{\gamma-1}{\gamma}} = 226.6 (25)^{\frac{0.4}{1.4}} \rightarrow \boxed{T_{3s} = 568.4 \text{ K}}$$

②

State 3 : $\eta_{\text{comp}} = \frac{\dot{W}_{\text{ideal}}}{\dot{W}_{\text{real}}} = \frac{-\dot{m} c_p (T_{3s} - T_2)}{-\dot{m} c_p (T_3 - T_2)}$

$$T_3 = T_2 + \frac{(T_{3s} - T_2)}{\eta_{\text{comp}}} = 226.6 + \frac{(568.4 - 226.6)}{0.9}$$

$$\boxed{T_3 = 606.38 \text{ K}}$$

①

State 4 : combustor $\rightarrow \dot{m} (\Delta h + \Delta ke + \cancel{\Delta pe}) = \dot{Q} - \dot{W}$

$$\dot{m} c_p (T_4 - T_3) = \dot{Q}$$

$$T_4 = T_3 + \frac{\dot{Q}}{\dot{m} c_p} = 606.38 + \frac{200 \times 10^6}{(680)(1001)}$$

$$\boxed{T_4 = 900.2 \text{ K}}$$

②
State 5: Turbine drives compressor $\dot{W}_{\text{turb}} = -\dot{W}_{\text{comp}}$

$$\Rightarrow -\dot{m} c_p (T_5 - T_4) = -[-\dot{m} c_p (T_3 - T_2)]$$

$$T_5 = T_4 - (T_3 - T_2)$$

$$= 900.2 - (606.38 - 226.6)$$

$$\boxed{T_5 = 520.4 \text{ K}}$$